

INSERT Data

Learning Objectives

- Write INSERT INTO statements to add single records to a table
- Insert multiple rows using a single INSERT command
- Handle common errors when inserting data into tables

Engage (5-7 minutes)

A new student joins the school. The clerk adds their details to the register—one row with name, class, and parent information. In SQL, this action is performed using the INSERT INTO command.

Explore (10-12 minutes)

Using a sample Students table, write INSERT statements to add five new students with varied data types—names, dates, and grades. Then attempt to insert a record with a duplicate primary key and observe the error message produced.

Explain (10-12 minutes)

INSERT INTO adds new records to a table. Specify the table name and columns, then provide values in matching order. You can insert multiple rows in one command by listing value sets separated by commas. Errors occur when you violate constraints like primary key uniqueness, NOT NULL requirements, or mismatched data types.

Elaborate (8-10 minutes)

A library database has a Books table. Write INSERT statements to add three books with ISBN (primary key), title, author, and publication year. Then explain what error would occur if you try to insert a book with an ISBN that already exists and how to prevent this.

Evaluate (5-8 minutes)

Write INSERT statements to add two records to a “Courses” table with fields: CourseID, CourseName, Credits, and Department. Explain one scenario where an INSERT would fail and how to handle it.